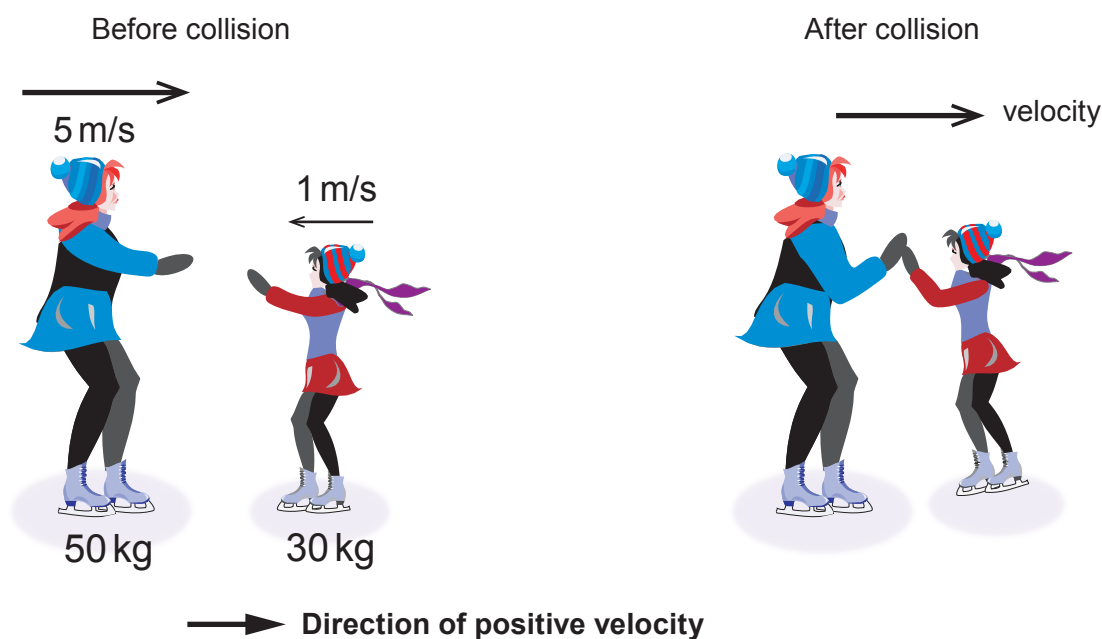


3. An ice skater of mass 50 kg and travelling with a velocity of 5 m/s to the right collides with a skater of mass 30 kg travelling with a velocity of 1 m/s **to the left**.



- (a) Use an equation from page 2 to calculate the total momentum of the skaters **before** the collision. [3]

Total momentum = kg m/s

- (b) After they collide the skaters move off together with a common velocity to the right. Use the equation:

$$\text{velocity} = \frac{\text{momentum}}{\text{mass}}$$

to calculate their velocity.

[2]

Velocity = m/s

